

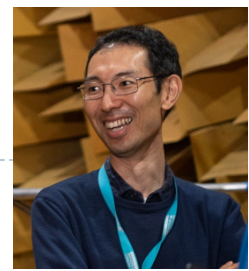
MECHENG726: Acoustics for Engineers

Chapter 1: Introduction (ver.1.0)

Section 0: Housekeeping

Who am I?

or "use case" = Yusuke's



- ▶ Yusuke Hioka (how to pronounce?: you-sca-y, he-oh-ka)

▶ Qualifications:

- ▶ 2000 BE, Dept System Design Engineering, Keio University, Japan.
- ▶ 2002 ME, Keio University, Japan.
- ▶ 2005 PhD in Engineering, Keio University, Japan.
Major: Signal Processing



▶ Professional Background

- ▶ 2005 – 2012 Research Engineer, NTT Laboratories, Tokyo
- ▶ 2010 – 2011 Visiting Researcher, Victoria University of Wellington
- ▶ 2013 – 2014 Lecturer in Dept. ECE, University of Canterbury
- ▶ 2014 – current Associate Professor in Dept. MME



▶ 3

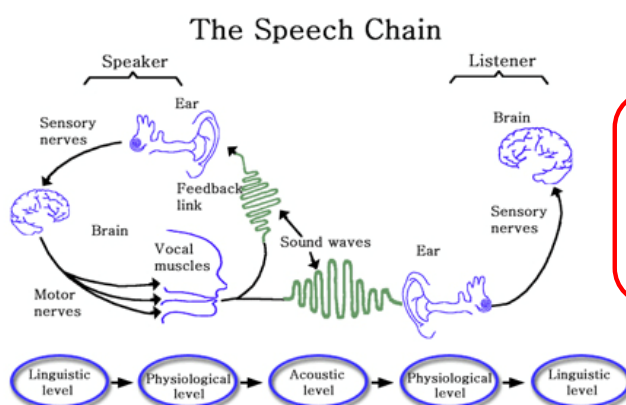
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Who am I? (cont'd)



▶ Research Area

Communication Acoustics which is concerned about everything to do with **communication by sound**



Traditional sub-disciplines included in communication acoustics

- Electro-acoustics and audio/speech signal processing
- Human auditory perception
- Psychology of hearing (psychoacoustics)
- Speech science and linguistics

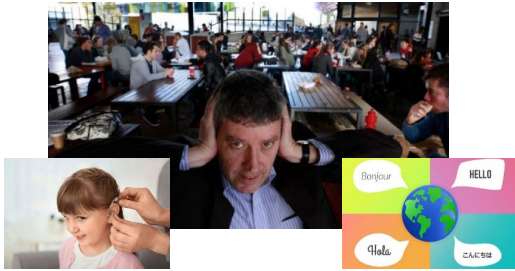
YH's part of this course will look into these areas

▶ 4

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Denes, Peter B., and Pinson, Elliot N. The Speech Chain : The Physics and Biology of Spoken Language. New York, N.Y.: W.H. Freeman, 1993. Print.

Research at CAL



Communication assistance for vulnerable listeners



Acoustics virtual reality (VR)



UAV audition



Bird call localisation and identification

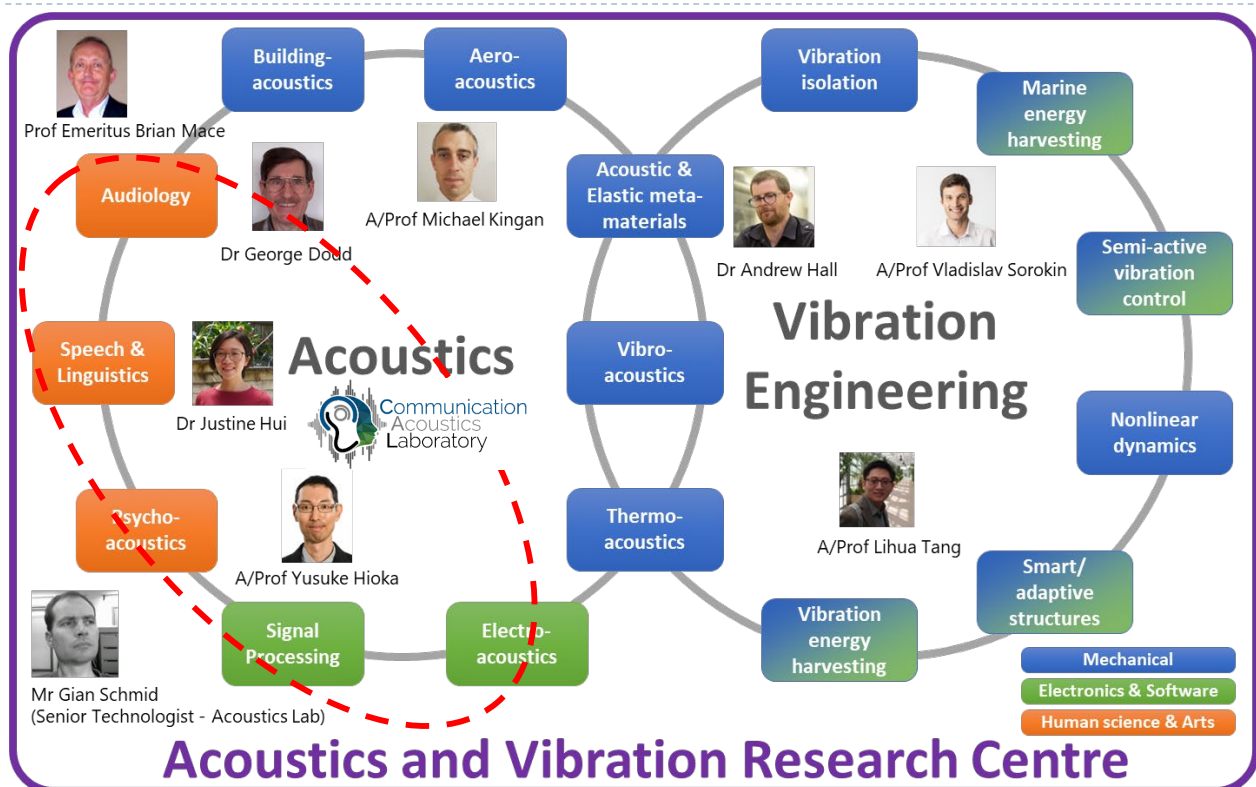


Speech privacy protection



Audio quality enhancement for telecommunication

Acoustics & Vibration Research Centre



Acoustics Lab

- ▶ Located in B422 in City Campus
- ▶ Hub of the Acoustics & Vibration Research Centre
- ▶ World-class chambers for acoustics research
 - ▶ Complying with relevant ISO standards for the measurement of sound level, absorption, transmission loss and impact insulation
 - ▶ Also used for commercial testing services
- ▶ We will visit the facilities in the Lab practical



Reverberation chamber



Anechoic chamber



Listening room

▶ 7

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Why acoustics?

- ▶ Me in Year4 of undergraduate...



▶ 8

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Contact information

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- ▶ Email: y.hioka@auckland.ac.nz
- ▶ Tel: 09 373 7599 (ext 81421)
- ▶ Office hours: by appointment only (email is the best way to contact me; please don't walk in randomly)
- ▶ For more details, please visit <https://profiles.auckland.ac.nz/y-hioka>

Course syllabus

- ▶ Lecturers
 - ▶ A/Prof Yusuke Hioka: y.hioka@auckland.ac.nz
 - ▶ A/Prof Michael Kingan (**Course coordinator**): m.kingan@auckland.ac.nz
 - ▶ Dr Andrew Hall: a.hall@auckland.ac.nz
- ▶ Lectures (also see course assessment planner on Canvas)
 - ▶ Wed: 1-3pm, Fri: 10-12am
 - ▶ YH: Weeks 1-5, MK: Weeks 7, 8,10, AH: 11, 12
- ▶ Assessments (also see course assessment planner on Canvas)
 - ▶ Assignment (**30%**): YH
 - ▶ Report (25%)
 - ▶ Pre-/Post-assignment quizzes (5%)
 - ▶ Lab practical (0%; BUT measurements from the lab needed for report)
 - ▶ Test (20%): YH
 - ▶ Final exam (**50%**): MK/AH

Assessments in YH's part

Test (20% of whole course grade)

- ▶ Held on Wed 16 September 1-2pm (week 7)*
- ▶ Exam will NOT cover YH's part
- ▶ Details to be announced in due course

*Subject to departmental approval

Assessments in YH's part

Assignment (30% of whole course grade)

- ▶ Components
 - ▶ 2.5% PRE-assignment quiz (during lecture slot Fri 28 August; NOT graded, attendance mark)
 - ▶ 25% Report (5pm Thurs 24 September)
Take home experiments to measure various acoustical parameters using a microphone, loudspeaker, and software.
 - ▶ 2.5% POST-assignment quiz (5pm Thurs 24 September; graded)
- ▶ Details about Assign report to be released in week 6.

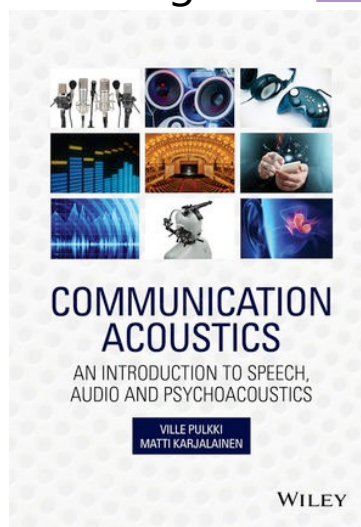
Assessments in YH's part

Lab practical

- ▶ One of streams in week 5 or 6 (ignore SSO timetable for labs).
Follow Canvas announcement for signing-up process.
- ▶ Held in Acoustics Laboratory (in Building 422).
- ▶ There is no grade for the lab, however you will be asked to attach/use your measurements from the lab in your Assignment report.
- ▶ Some tasks in the lab are designed to help you to do the Assignment.

Recommended textbook (YH's part)

- ▶ Ville Pulkki, Matti Karjalainen, *Communication Acoustics: An Introduction to Speech, Audio and Psychoacoustics*, Wiley 2015, ISBN: 978-1-118-86654-2
- ▶ **Online copy available** through the [library web site](#)



Section 1: Introduction to acoustics

15

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What is acoustics?

- ▶ How do you define “acoustics”?
- ▶ Let's check its definition on dictionaries...
 - ▶ *the study of **sound*** (McMillan)
 - ▶ *the **scientific study** of **sound*** (Cambridge/Longman/Oxford)
 - ▶ *the **scientific study** of **sound and sound waves*** (Collins, dictionary.com)
 - ▶ *a **science** that deals with the production, control, transmission, reception, and effects of **sound*** (Merriam-Webster)
- ▶ Origin of word: Greek *akouein* (= to hear)

So what is sound then?

- Q. If a tree falls in the forest and no one is there, does it make a sound?



► 17

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Objective sound vs Subjective sound

- Let's check a dictionary again for the word "sound"...
- 1. A sound is something that you can hear.
- 2. Sound is energy that travels in waves through air, water, or other substances, and can be heard.

Definition of 'sound'

sound

noun and verb uses

Collins COBUILD

(saʊnd  )

Word forms: plural, 3rd person singular present tense **sounds** , present participle **sounding** , past tense, past participle **sounded** 

1. COUNTABLE NOUN

A sound is something that you hear.

*Peter heard the sound of gunfire.
Liza was so frightened she couldn't make a sound.
There was a splintering sound as the railing gave way.
...the sounds of children playing.*

Synonyms: noise, racket, din, report [More Synonyms of sound](#)

2. UNCOUNTABLE NOUN

Sound is energy that travels in waves through air, water, or other substances, and can be heard.

The aeroplane will travel at twice the speed of sound.

3. SINGULAR NOUN

The sound on a television, radio, or CD player is what you hear coming from the machine. Its loudness can be controlled

Word Frequency 

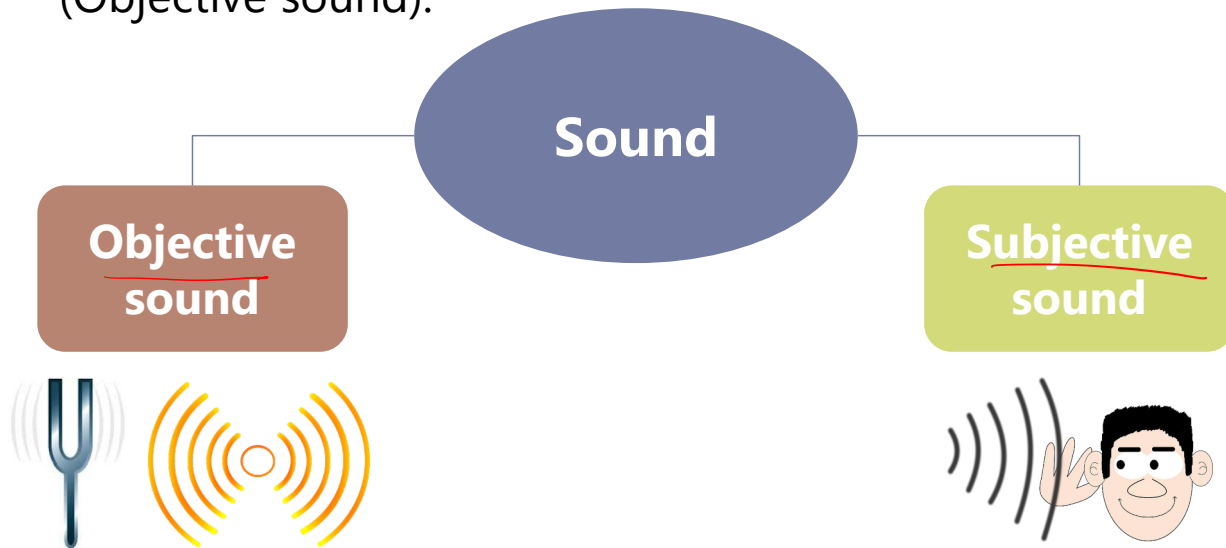

(Collins)

► 18

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Objective sound vs Subjective sound

- ▶ The noun “sound” in English has two different meanings.
- ▶ One refers to the **subjective percept** (Subjective sound) and another refers to the **physical phenomenon** (Objective sound).



We may answer the question

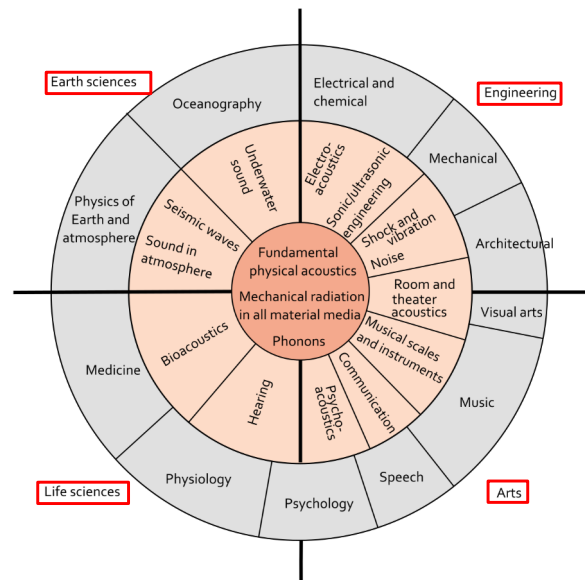
- ▶ Q. *If a tree falls in the forest and no one is there, does it make a sound?*



- ▶ A model answer. A falling tree causes a **physical sound event** (objective sound) that can be recorded and analysed every time, but it does not make any **sound in the sense of an auditory perception** (subjective sound) if there is no subject to hear the event. (Pulkki Chapt 2)

Revisiting “acoustics”

- ▶ There are many different fields of acoustics for study and practice as you can see in Lindsay’s *Wheel of Acoustics*.
- ▶ The “Wheel of Acoustics” was created by R. Bruce Lindsay for *The Science of Acoustics*, published in the *Journal of the Acoustical Society of America*.
- ▶ This wheel shows the fields of acoustics starting with the four broad fields of **Earth Sciences**, **Engineering**, **Life Sciences**, and the **Arts**.



“Wheel of Acoustics” by Robert Bruce Lindsay

<https://en.wikipedia.org/wiki/Acoustics>

(From “Field of Acoustics” <https://exploresound.org/what-is-new/fields-of-acoustics/>)

Subdisciplines in acoustics

- ▶ Acoustical Society of America (ASA) is the largest research institute in acoustics.

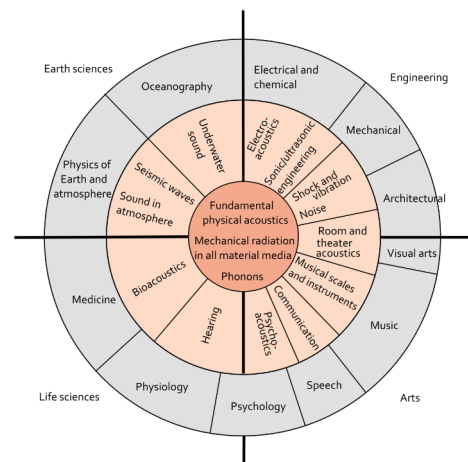


Acoustical Society of America

The purpose of the Society is to generate, disseminate, and promote the knowledge and practical applications of acoustics.

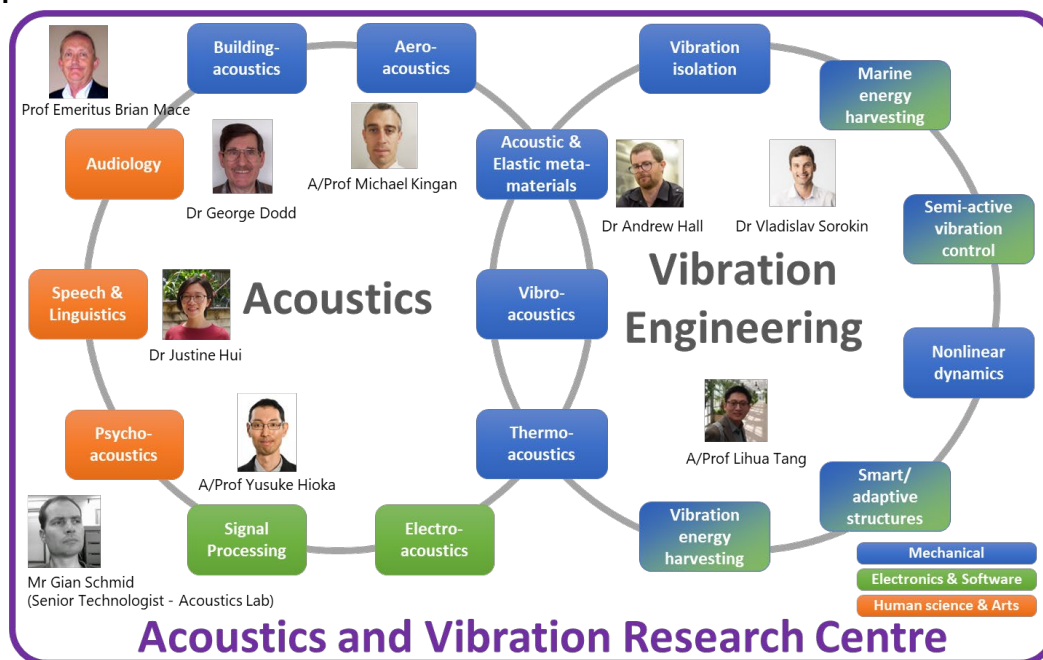
- ▶ ASA has 13 technical committees spanning across the wider subdisciplines listed in the Wheel of Acoustics.

- Acoustical Oceanography
- Animal Bioacoustics
- Architectural Acoustics
- Biomedical Acoustics
- Engineering Acoustics
- Musical Acoustics
- Noise
- Physical Acoustics
- Psychological and Physiological Acoustics
- Signal Processing in Acoustics
- Speech Communication
- Structural Acoustics and Vibration
- Underwater Acoustics



Subdisciplines in acoustics

- ▶ Acoustics and Vibration Research Centre also has expertise of different fields in acoustics.



▶ 23

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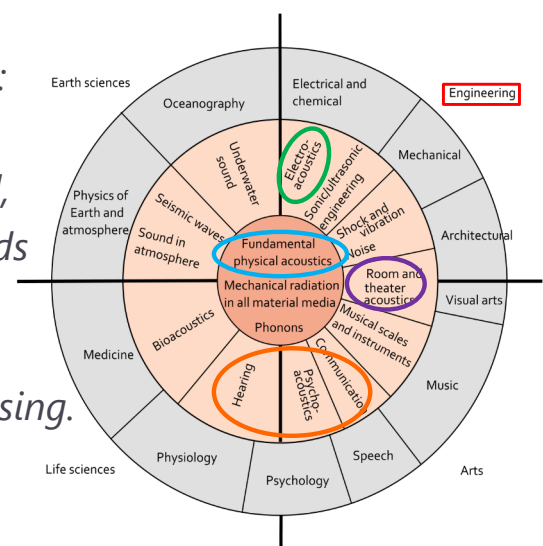
Aim of this course

- ▶ Course overview (in syllabus)

*This course is intended to prepare students for work as an **acoustic engineer**. The syllabus includes(snip) with a strong emphasis on teaching knowledge which students will be able to apply in a career as an **acoustic engineer** (snip).*

The major topics covered in this course are:

- physical acoustics,
- human hearing and perception of sound,
- ⊕ legal requirements and standard methods for quantifying sound,
- building acoustics,
- electroacoustics and audio signal processing.

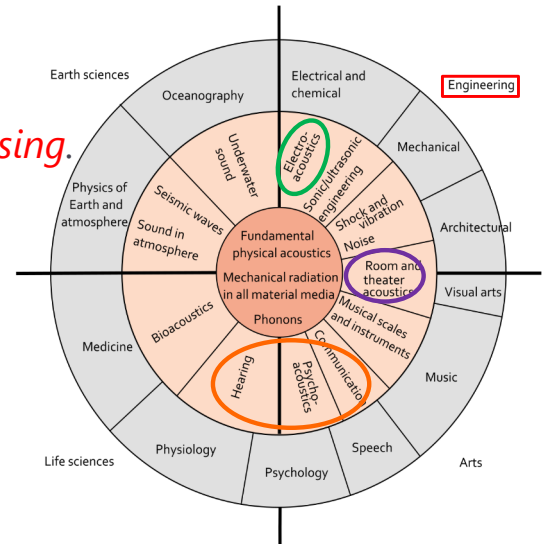


▶ 24

MECHENG726 2026 Chap1 Introduction

Aim of this course

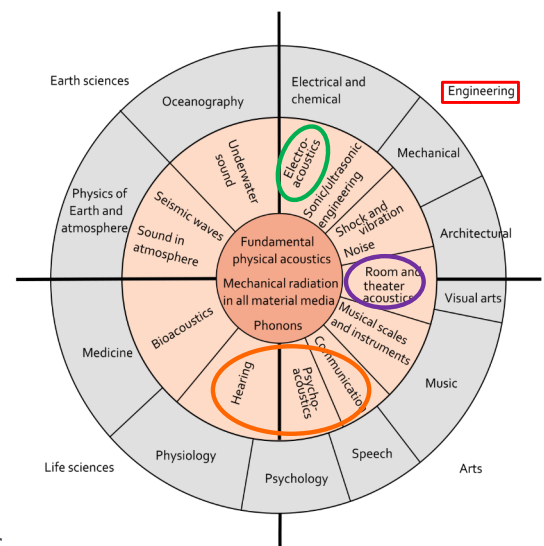
- ▶ YH's part will mainly cover the topics highlighted in red
 - *physical acoustics,*
 - *human hearing and perception of sound,*
 - *legal requirements and standard methods for quantifying sound,*
 - *building acoustics,*
 - *electroacoustics and audio signal processing.*



Strong emphasis on **electro-acoustics** and **signal processing**

Structure of YH's part

- ▶ Chapter2: Signal processing for acoustics
 - ▶ Digital signal acquisition
 - ▶ Spectral analysis
 - ▶ Impulse response & Frequency response (transfer function)
 - ▶ Digital filters
- Chapter3: Science of hearing
 - ▶ Physiology of human hearing
 - ▶ Basic function of hearing
 - ▶ Psychoacoustics
 - ▶ Spatial acoustics
- Chapter4: Acoustic measurement & analysis
 - ▶ Sound level measurement
 - ▶ Impulse response measurement
 - ▶ Room acoustics property measurement
- ▶ Chapter5: Microphone & loudspeakers
 - ▶ Microphones: types & properties
 - ▶ Loudspeakers: types & properties
 - ▶ Measuring microphones and loudspeakers
 - ▶ Advanced techniques using microphones & louds

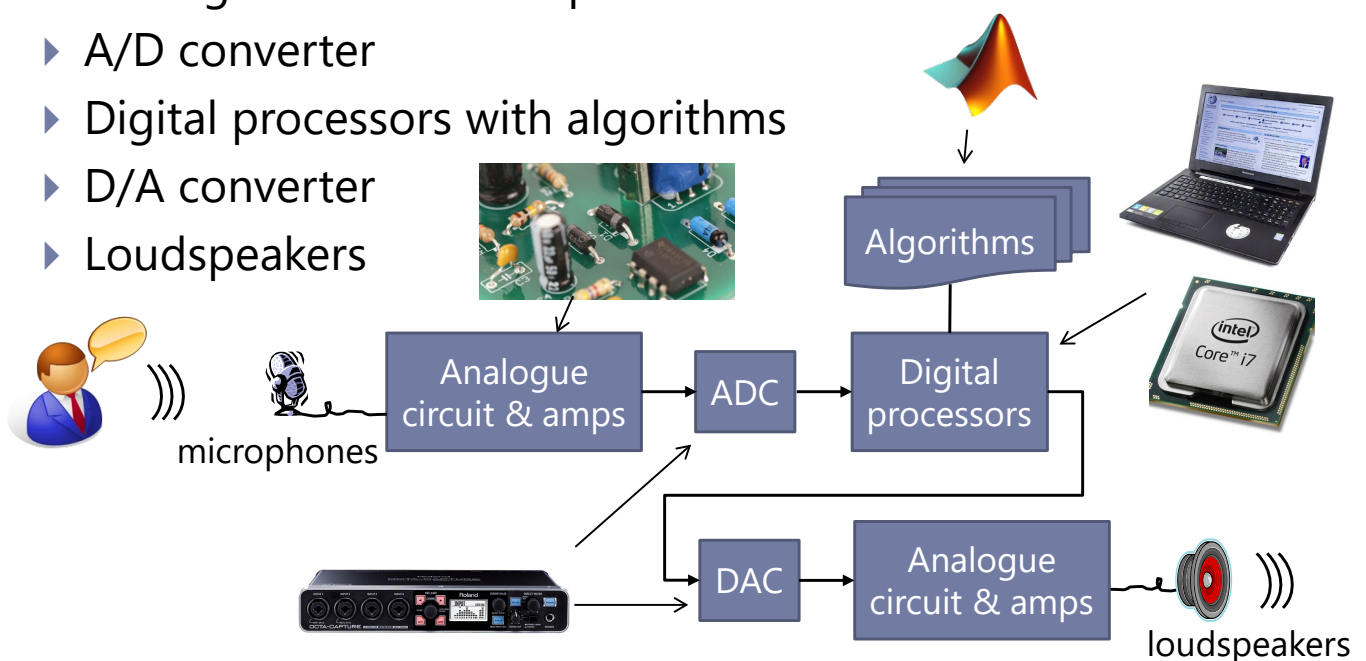


Electro-acoustics & audio signal processing

- ▶ **Electro-acoustics (EA)** is a sub-discipline of acoustics.
- ▶ Originally it referred to technologies for conversion between sound waves and electrical signals.
- ▶ EA is concerned about technologies to **record, play, analyse, and process** sound using electronics devices.
- ▶ Topics typically covered by EA are:
 - ▶ Microphones (**Recording**)
 - ▶ Loudspeakers (**Playing**)
 - ▶ Measurement & analysis (**Signal processing**)
 - ▶ Analogue circuits
 - ▶ Digital signal processing
- ▶ These techniques are also essential for other areas of acoustics.

Devices/equipment involved in EA

- ▶ Microphones
- ▶ Analogue circuits & amplifiers
- ▶ A/D converter
- ▶ Digital processors with algorithms
- ▶ D/A converter
- ▶ Loudspeakers



Applications using EA technologies

- ▶ Most modern **digital devices** that come with functions to **record/play sound** use EA technologies.



Telephone
/smartphones



Portable audio devices
(CD, iPod, etc.)



Hearing aids /
Noise cancelling
headphones



Public address system



Electric
instruments

Applications using EA technologies

- ▶ Another key application of EA technologies is **acoustic measurement** of rooms and audio devices.



Room acoustics measurement



Measuring a loudspeaker/microphone in an anechoic chamber.

YH's part learning objectives

By the end of YH's part, students will understand

- ▶ How to calculate a spectrum of sound signal
- ▶ How to describe the behaviour of an object in acoustics
- ▶ Different types of digital filters and their characteristics
- ▶ Basic structure and mechanism of human auditory perception
- ▶ How to measure the impulse response of an object
- ▶ Various acoustical properties of rooms and techniques for their measurements and evaluations
- ▶ Different types of microphones/loudspeakers and their properties
- ▶ How to measure the properties of microphones and loudspeakers
- ▶ How the techniques in EA are used in practical applications

YH's part learning objectives (cont'd)

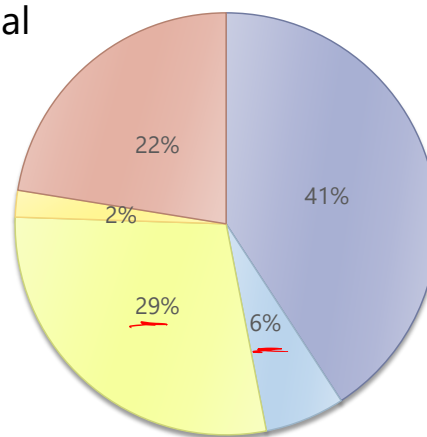
For all:

Acquire fundamental skills and knowledge for **recording, playing and analysing** sound as well as **measuring acoustical properties** of objects informed by basic knowledge about **science of hearing**.

For ME students:

Understand fundamental concepts of electronics and signal processing essential for acoustics engineer through **practical problems in acoustics**.

Class Demographics



For MT/ECSE students:

Understand how the techniques covered in previous courses of electronics and signal processing are applied to **practical problems in acoustics**.

Previous relevant courses

- ▶ **Mechanical:**

- ▶ MECHENG 322: Control Systems
- ▶ MECHENG 722: Engineering Vibrations

- ▶ **Mechatronics:**

- ▶ MECHENG 322: Control Systems
- ▶ MECHENG 370: Electronics and Signal Processing
- ▶ MECHENG 705: Mechatronics Systems

- ▶ **Electrical/Computer systems/Software**

- ▶ ELECTENG 331: Signals and Systems
- ▶ ELECTENG 332: Control Systems